4. Factorization into A = LU

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1. Inverse of a Product

How do we find the inverse of a matrix product, $(AB)^{-1}$?
If we know A and B are invertible, we apply the operations in **reverse order**.

$$(AB)^{-1} = B^{-1}A^{-1}$$

Proof:

$$(AB)(B^{-1}A^{-1}) = A(BB^{-1})A^{-1} = A(I)A^{-1} = AA^{-1} = I$$

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2. Inverse of a Transpose

A major property used in future lectures: The inverse of a transpose is the transpose of the inverse. The order of operations does not matter.

$$(A^T)^{-1} = (A^{-1})^T$$

Derivation:
Start with $AA^{-1} = I$.
Transpose both sides (remembering that for products, transposing reverses the order):

$$(A^{-1})^T A^T = I^T = I$$

Therefore, $(A^{-1})^T$ must be the inverse of A^T.

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3. A = LU Factorization (The Core Concept)

This is the matrix description of **Gaussian Elimination**. Ideally, we want to factor matrix A into a Lower triangular matrix (L) and an Upper triangular matrix (U).

Assumption: No row exchanges are required (pivots are non-zero).

2x2 Example

Let Matrix $A = \begin{bmatrix} 2 & 1 \\ 8 & 7 \end{bmatrix}$.

Step 1: Elimination
We need to eliminate the 8 in position $(2, 1)$.

- Pivot: 2
- Multiplier: 4 ($2 \times 4 = 8$)
- Operation: Row 2 - 4(Row 1).

The **Elementary Matrix** (E_{21}) that performs this subtraction is:

$$E_{21} = \begin{bmatrix} 1 & 0 \\ -4 & 1 \end{bmatrix}$$

Applying this to A:

$$E_{21}A = \begin{bmatrix} 1 & 0 \\ -4 & 1 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ 8 & 7 \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ 0 & 3 \end{bmatrix} = U$$

Here, U is the **Upper Triangular** result of elimination.

Step 2: Factorization (Inverting the Operation)
We want to express A in terms of U. We move E_{21} to the other side by using its inverse.

$$A = E_{21}^{-1}U$$

$$A = LU$$

The inverse of subtracting 4 times row 1 is **adding 4 times row 1**.

$$L = E_{21}^{-1} = \begin{bmatrix} 1 & 0 \\ \mathbf{4} & 1 \end{bmatrix}$$

Key Insight

The multiplier used in elimination (4) shows up directly in the L matrix at the exact position $(2, 1)$ where we eliminated the entry.

The LDU Factorization (Variation)

Sometimes we want to separate the pivots from U. We can split U into a diagonal matrix D (containing the pivots) and a new U (with 1s on the diagonal).

$$A = LDU$$

$$\begin{bmatrix} 2 & 1 \\ 8 & 7 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 4 & 1 \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} 1 & 1/2 \\ 0 & 1 \end{bmatrix}$$

4. The 3x3 Case (Why L is better than E)

For a 3x3 matrix, elimination usually involves three steps (matrices E_{21}, E_{31}, E_{32}).

The Elimination View (Messy):

$$E_{32}E_{31}E_{21}A = U$$

If we multiply all the E's together to get one matrix E, the numbers mix up and become complicated. The history of operations is tangled.

The Factorization View (Clean):
We move the E's to the right side by inverting them. Note the order reverses.

$$A = (E_{21}^{-1}E_{31}^{-1}E_{32}^{-1})U$$

$$A = LU$$

Visualizing L:
If no row exchanges are needed, L is formed by placing the **multipliers** directly into their respective positions below the diagonal.

$$L = \begin{bmatrix} 1 & 0 & 0 \\ l_{21} & 1 & 0 \\ l_{31} & l_{32} & 1 \end{bmatrix}$$

Deep Dive: The "Magic" of L vs E (Concrete Example)

This is a subtle point. Matrix multiplication usually implies mixing, but L keeps multipliers distinct while E mixes them.

1. The "Messy" Way (E):
Imagine subtracting **2** times Row 1 from Row 2, then **5** times Row 2 from Row 3.

$$E_{total} = E_{step2} \times E_{step1} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & \mathbf{-5} & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ \mathbf{-2} & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ \mathbf{10} & -5 & 1 \end{bmatrix}$$

Where did the 10 come from? Row 3 is affected by Row 1 *indirectly* through Row 2. The operations compound ($(-2) \times (-5) = 10$). The history is tangled.

2. The "Clean" Way (L):
L reverses the order and inverts the signs to **reconstruct** A.

$$L = E_{step1}^{-1} \times E_{step2}^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ \mathbf{2} & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & \mathbf{5} & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ \mathbf{2} & 1 & 0 \\ 0 & \mathbf{5} & 1 \end{bmatrix}$$

Where is the 10? It's gone! The multipliers **2** and **5** sit purely in their positions (l_{21} and l_{32}).

Intuition:

- E **(Elimination)** records the **process** (affected by changing rows).
- L **(Reconstruction)** records the **instruction set** ("Row 3 needed 5 of Row 2").

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5. Computational Complexity (Operation Counts)

How expensive is elimination for an $n \times n$ matrix?
Operation definition: 1 Multiply + 1 Subtract.

Cost for Matrix A (Left Side)

- **Step 1:** The first pivot row modifies $n-1$ rows below it. Each row has n elements (approx). Operations $\approx n^2$.
- **Step 2:** The next pivot modifies $n-2$ rows. Operations $\approx (n-1)^2$.
- **Sum:** $n^2 + (n-1)^2 + \cdots + 1^2 \approx \sum_{i=1}^{n} i^2$.

Using the calculus approximation ($\int x^2 dx = \frac{1}{3}x^3$):

$$\text{Operations for A} \approx \frac{1}{3}n^3$$

Cost for Vector b (Right Side)

We also have to process the right-hand side vector b.

- Operations $\approx n^2$.

Practical Implication

For very large n (e.g., $n = 1000$):

- Factorization ($A = LU$) is expensive ($\frac{1}{3}n^3$).
- Solving for specific solutions (b) is cheap (n^2).
  Once you pay the cost to find L and U, you can solve for many different b's very quickly.

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6. Permutations (Row Exchanges)

If a 0 appears in a pivot position, we must swap that row with a row below it.

- **Permutation Matrix (P):** The identity matrix with rows reordered.

- To swap rows i and j of matrix A, compute PA.

Properties of Permutations:

1. There are $n!$ permutation matrices of size $n \times n$.
2. They form a **Group**:
   - Product of two P's is a P.
   - Inverses exist (and are P's).
3. **Crucial Property:** The inverse is the transpose.

$$P^{-1} = P^T$$

General Factorization:
If row exchanges are required, the factorization becomes:

$$PA = LU$$

(Where P is the permutation matrix that puts the rows in the correct order for elimination).